Sahaj Bhadja

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Summary

Backend Software Engineer pursuing an MSCS at USC with 2+ years building **scalable platform services**, **public APIs**, and **async orchestration and ETL pipelines** across distributed systems, using **AI coding assistants** for fast, reliable delivery.

Education

University of Southern California, Los Angeles, CA — **August 2024 – May 2026**
Master of Science in Computer Science — *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India — **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering — *GPA: 3.88/4*
Coursework: Operating Systems, Database Systems, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife — **Los Angeles, CA**
Software Engineer — *September 2025 – Present*
- Built **fault-tolerant async orchestration** with explicit job-state transitions and failure isolation across 5K+ monthly jobs, sustaining 99%+ reliability and improving throughput **3x** under concurrent load on a multi-tenant platform.
- Designed **multi-tenant backend services and APIs** for 500+ users with normalized SQL schemas and row-level isolation, optimizing query patterns to cut latency from 1.2s to 350ms for scalable, reliable performance.

Grad8 — **Los Angeles, CA**
Software Engineer Intern — *June 2025 – August 2025*
- Built and maintained **public-facing backend APIs** (Flask, REST) handling 10K+ daily requests, integrating third-party providers behind a switchable architecture with CI/CD releases and clear code-review workflows.
- Leveraged **AI coding assistants and LLMs** to accelerate prototyping, testing, and refactoring while shipping reliable production features for 150+ users in a fast-paced startup environment.

Aasma Technology Solutions — **India**
Software Developer Engineer - (SecureSync) — *August 2023 – July 2024*
- Engineered **ETL pipelines and microservices** with AWS Glue and S3, automating data movement and improving pipeline efficiency **68%** while supporting distributed, cross-platform production workloads with reliable scheduling.
- Maintained **CI/CD pipelines and 20+ automated test suites** (Jenkins, PyTest, Mockito), increasing reliability through code reviews, debugging, and version-control workflows across coordinated service releases.

Aasma Technology Solutions — **India**
Full Stack Developer Intern - (VibrantChem) — *May 2023 – August 2023*
- Developed **REST APIs and microservices** (Node.js, Express) with WebSocket real-time updates and alerting across eight workflows, syncing 2M+ records via Azure SQL for reliable, scalable data flow.
- Automated **Docker-based deployments** via Jenkins CI/CD with Sentry and Azure Monitor for observability and alerting, cutting release time **50%** and improving platform stability and incident response.

LST Capitals — **India**
Web Developer Intern — *November 2022 – January 2023*
- Built **Flask APIs with JWT and wallet-based authentication** plus billing and payment workflows across 50K+ records, integrating third-party gateways with real-time sync and sub-200ms responses.
- Automated **billing and refund workflows** through idempotent webhook handling, reducing manual handling **70%**, and documented 12+ backend integration points for client and stakeholder alignment.

Skills

Languages: Python, Java, JavaScript/TypeScript, SQL, C++, Bash
Backend & APIs: FastAPI, Flask, Node.js, REST APIs, Public APIs, Microservices, Authentication (JWT), Webhooks
Platform & Orchestration: Workflow Orchestration, Async Job Scheduling, ETL Pipelines, Distributed Systems, Reliability, Scalability
Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Azure SQL, Query Optimization
Cloud & Delivery: AWS (S3, Glue, ECS, RDS), Azure, Docker, Kubernetes, CI/CD, Jenkins, Terraform
Practices & AI: AI Coding Assistants (Claude, Cursor), Code Reviews, Testing, Version Control, Debugging, Agile

Projects

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Built a **containerized REST API** on AWS (CodePipeline, ECS/Fargate) with automated CI/CD and validation gates, delivering production-grade reliability and scalability across data-processing workflows.

Big-Brain Live | *Cloud Data Platform, REST APIs, GCP*
- Built a **cloud data platform** with an ingestion and indexing pipeline over 50K+ records, adding incremental updates and performance tuning to cut retrieval latency **40%** through REST APIs.

FinNexus Live | *Next.js, Kubernetes, MongoDB, Plaid*
- Built a **scalable platform** handling 1M+ transactions for 50K+ users with billing workflows, token-based authentication, real-time updates, and Kubernetes-backed CI/CD under high transaction volume.